

Project

Systems with Analog Integrated Circuits

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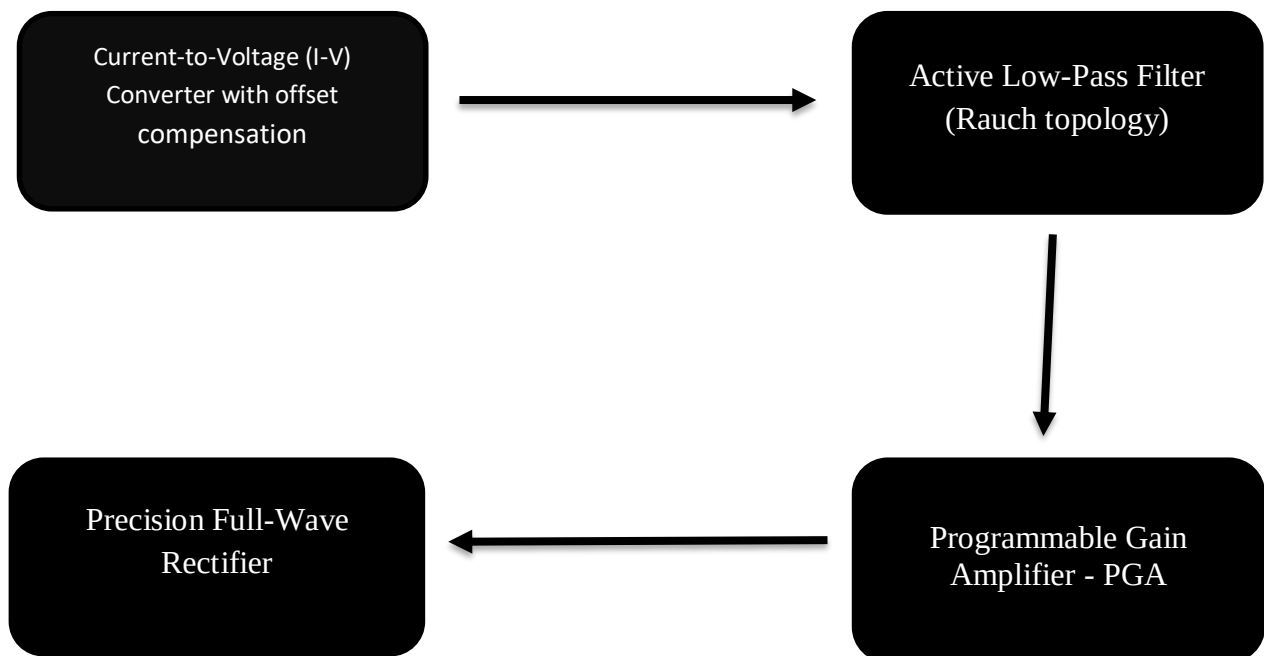
1. Project subject

1.1 Requirements

This project aims to design complete analog signal acquisition and conditioning chain, organized into four distinct functional stages:

- **Stage 1:** Current-to-Voltage (I-V) Converter with offset compensation
- **Stage 2:** Active Low-Pass Filter (Rauch topology)
- **Stage 3:** Programmable Gain Amplifier - PGA
- **Stage 4:** Precision Full-Wave Rectifier

1.2 Block Diagram



1.3 Specifications

1.3.1 Stage 1: Current-to-Voltage (I-V) Converter with offset compensation

The first stage of the system consists of a current-to-voltage (I-V) converter implemented using an LT1679 operational amplifier in an inverting configuration, equipped with a DC compensation network. This stage converts the input current signal into a proportional voltage signal.

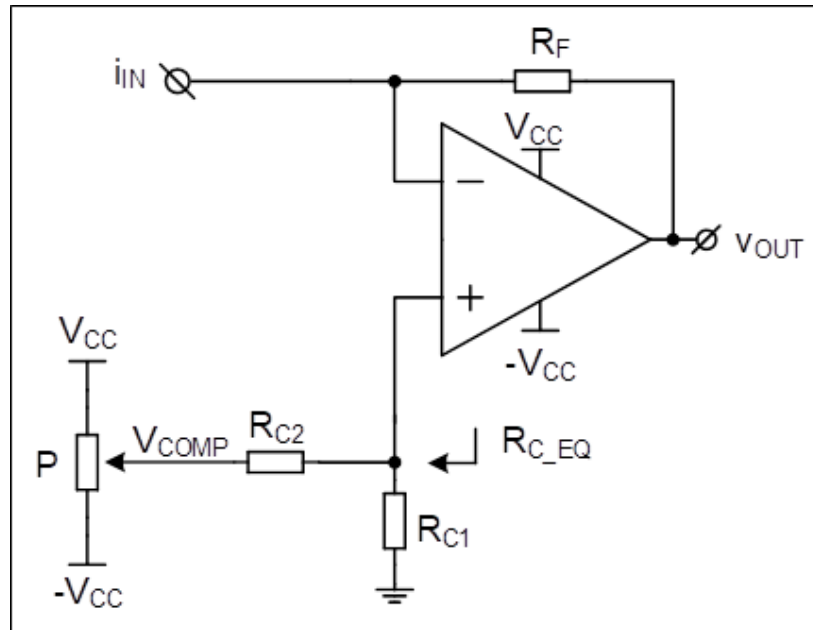


Figure 1. Current-to-Voltage (I-V) Converter with offset compensation

The input signal is supplied by a single-ended current source with the following specifications:

Minimum amplitude: $I_{IN,min} = 1,58 * 10^{-4} A = 158\mu A$

Maximum amplitude: $I_{IN,max} = 3,16 * 10^{-4} = 316\mu A$

Gain: 5000

1.3.2 Active Low-Pass Filter (Rauch topology)

The second stage of the signal-conditioning chain consists of a low-pass filter based on the Rauch (multiple-feedback) topology, implemented with a single operational amplifier.

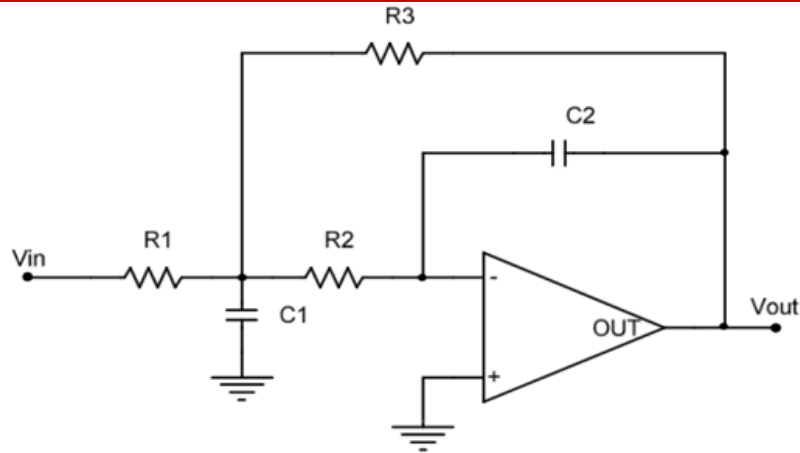


Figure 2. Active Low-Pass Filter (Rauch topology)

Passband linear gain: $|H_0| = 1$

Minimum input resistance: $R_{IN.min} = 1,5k\Omega$

Bandwidth: $BW = 9kHz$

Quality factor: $Q=1$

1.3.3 Programmable Gain Amplifier – PGA

Stage 3 is a Non-Inverting Programmable Gain Amplifier (PGA) utilizing the 'RG parallel' topology, which serves the critical role of signal scaling and dynamic range optimization for the entire acquisition chain. This configuration is characterized by a fixed feedback resistor (R_F) and a bank of multiple gain-setting resistors ($R_{G1}, R_{G2}, R_{G3}, R_{G4}$) connected in parallel. Each resistor is controlled by an independent switch (SW_1 to SW_4), allowing the user to select the desired amplification level based on the input signal intensity to ensure an optimal voltage range for the final stage.

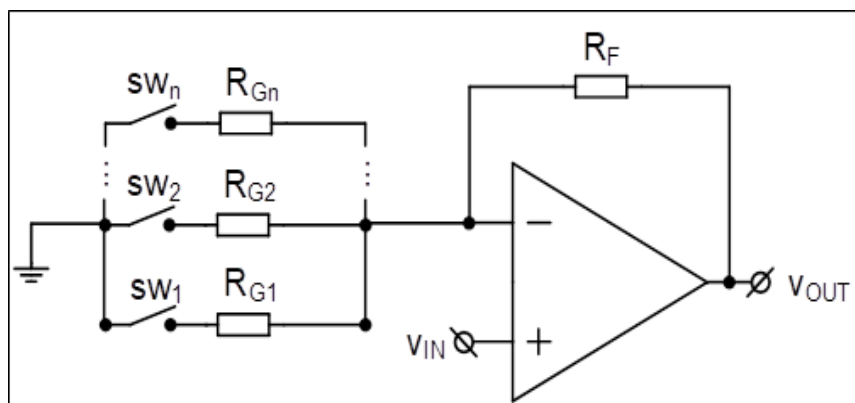


Figure 3. Programmable Gain Amplifier – PGA

Minimum linear gain: $A_{V,min} = 10dB$

Maximum linear gain: $A_{V,max} = 16dB$

Resolution (minimum step): 2dB

Number of steps: 4

1.3.4 Precision Full-Wave Rectifier

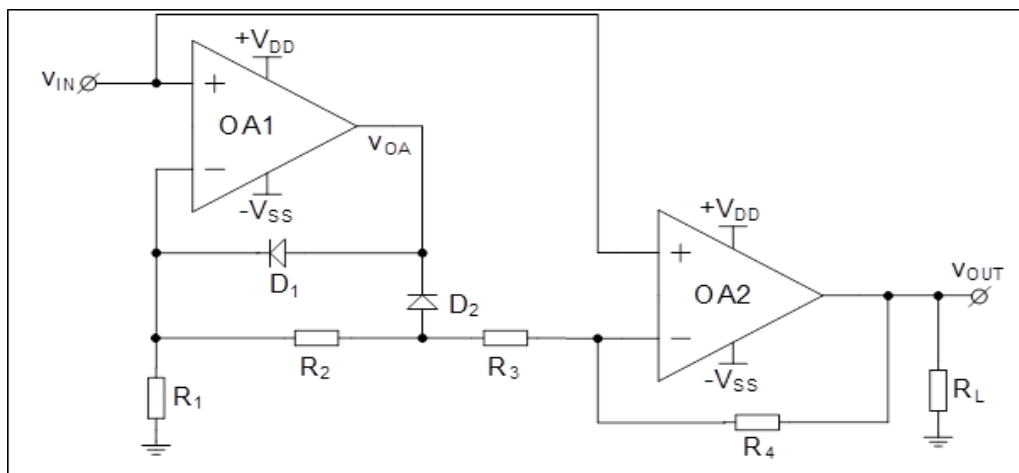


Figure 4. Precision Full-Wave Rectifier

Stage 4 functions as a precision full-wave rectifier, utilizing two operational amplifiers, two diodes, and a specialized resistive network to accurately convert any bipolar input signal into a strictly positive unipolar output while eliminating the voltage drop typically caused by standard diodes.

Linear gain: $A_V = 1$

2.Dimensioning

2.1 Stage 1

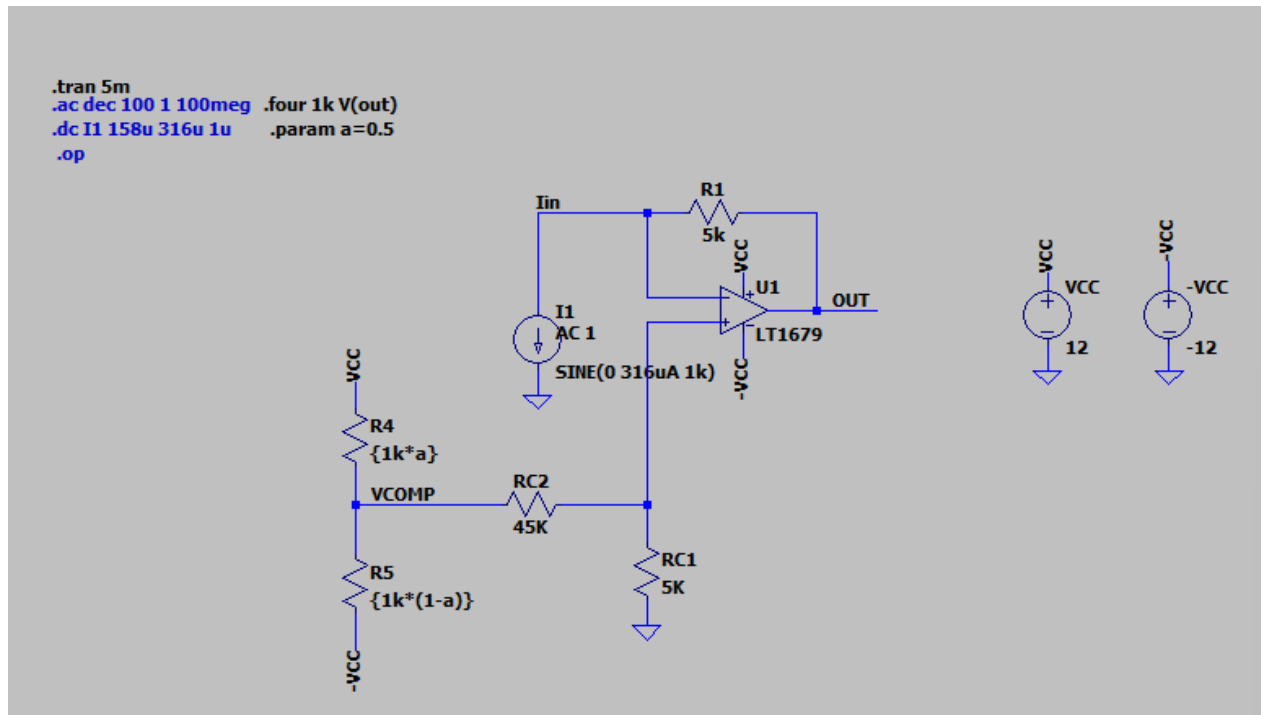


Figure 5. Stage 1

For the proper design of the current-to-voltage (I-V) converter, the first step was to evaluate the offset error introduced by the operational amplifier and the input bias currents. For this purpose, the following test configuration was adopted:

- the input current source was set to zero, $I_{IN} = 0$.
- the non-inverting input (V^+) was temporarily connected to ground.

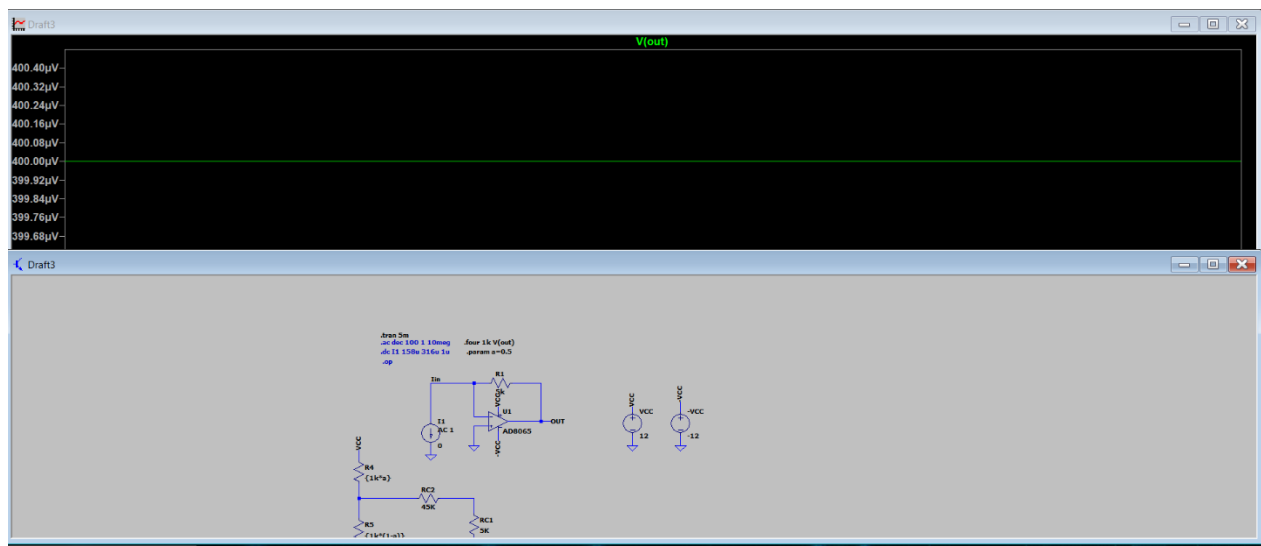


Figure 6. DC offset voltage error

The operating point analysis indicated a residual output voltage of 0.4mV, attributable to DC offset. A DC compensation network was therefore implemented at the non-inverting input (V^+) to inject a compensating voltage of about -0.4mV, ensuring that $V_{OUT} \approx 0V$ for $I_{IN} = 0$.

The output voltage is calculated using this formula: $V_{OUT} = -i_{IN} * R_1$

The formula for the gain of the system is : $A_V = \left| \frac{V_{OUT}}{I_{IN}} \right| = \left| \frac{-I_{IN} * R_1}{I_{IN}} \right| = R_1$, $A_V = 5000$
 $\Rightarrow R_1 = 5k\Omega$

Sizing R_{C1} and R_{C2} :

$$V^+ = V_{COMP} * \frac{R_{C1}}{R_{C1} + R_{C2}}$$

To enhance precision, a 1/10 scaling factor was applied to align the potentiometer's voltage range with the required offset correction.

$$0.4mV = V_{COMP} * \frac{1}{10} \Rightarrow V_{COMP} = 0.4mV * 10 = 4mV$$

$$\frac{R_{C1}}{R_{C1} + R_{C2}} = \frac{1}{10} \Rightarrow R_{C2} = 9 * R_{C1}$$

I chose $R_{C1} = R_1 = 5k\Omega$ to cancel the bias current errors and keep the circuit stable.

$$\Rightarrow R_{C2} = 9 * 5k\Omega = 45k\Omega$$

Main parameters used: $R_{POT} = 1k\Omega$

$V_{CC} = 12V$ and $-V_{CC} = -12V$ from LT1679 datasheet

I ran an operating point (.op) analysis to find the best setting for a . I found that when $a = 0.5$, the output voltage is almost exactly zero. I selected this value to ensure the DC offset is perfectly fixed.

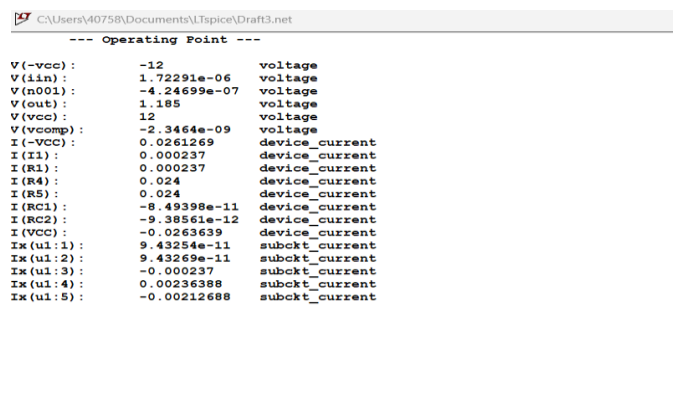


Figure 7. OP analysis

2.2 Stage 2

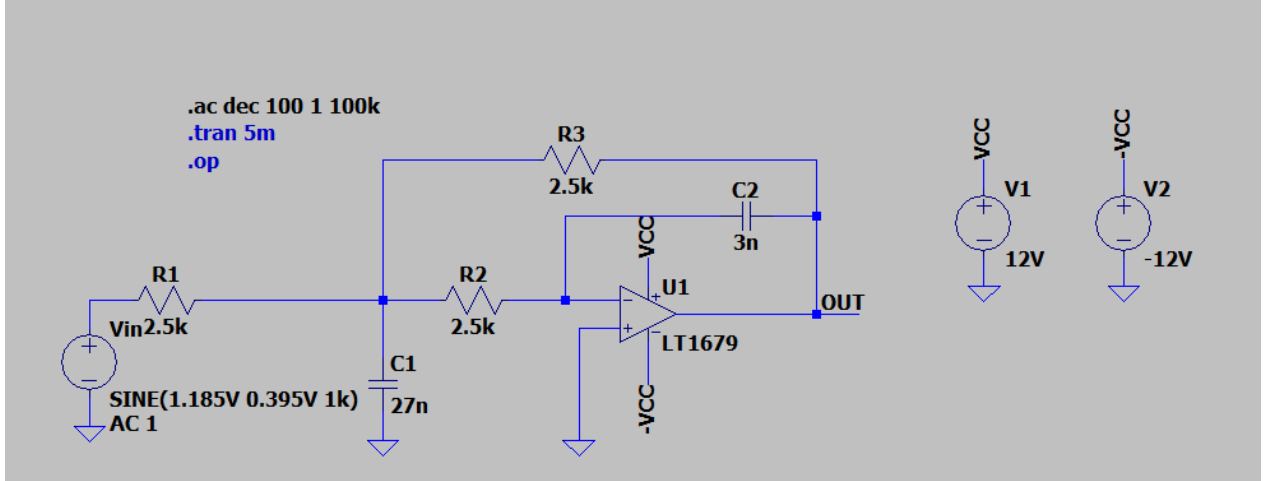


Figure 8. Implementation of stage 2: Rauch LPF

The second stage of the system is an active second-order low-pass filter using the Rauch (multiple-feedback) topology. The main purpose of this stage is to clean the signal by limiting the bandwidth to 9 kHz, removing high-frequency noise while keeping the signal linear.

The filter is designed for a unity passband gain ($|H_0| = 1$). In a Rauch filter, the gain is set by the ratio of the feedback resistor (R_3) to the input resistor (R_1):

$$|H_0| = \frac{R_3}{R_1}$$

By choosing $R_1 = R_3 = 2.5 \text{ k}\Omega$, I obtained the required gain:

$$|H_0| = \frac{2.5 \text{ k}\Omega}{2.5 \text{ k}\Omega} = 1 \frac{\text{V}}{\text{V}}$$

The cutoff frequency is set to $f_c = 9 \text{ kHz}$ and is calculated using R_2 , R_3 , C_1 , and C_2 :

$$f_c = \frac{1}{2\pi\sqrt{R_2 R_3 C_1 C_2}}$$

Using $R_1 = R_2 = R_3 = 2 \text{ k}\Omega$, combined with $C_1 = 27 \text{ nF}$ and $C_2 = 3 \text{ nF}$, the resulting frequency is approximately 8.84 kHz, which effectively satisfies the 9 kHz bandwidth requirement. The quality factor (Q) is specified as 1. For this Rauch topology, when all resistors are equal, the formula for Q simplifies to:

$$Q = \frac{\sqrt{C_1}}{\sqrt{C_2}} = \frac{\sqrt{27}}{\sqrt{3}} = \frac{\sqrt{9}}{3} = \frac{3}{3} = 1$$

This matches the project specifications.

The input resistance is approximately $R_{in} \approx R_1 = 2.5 \text{ k}\Omega$, which is higher than the required minimum of $1.5 \text{ k}\Omega$, ensuring that this stage does not overload the previous one. The LT1679 operational amplifier is used here, powered by a $\pm 12 \text{ V}$ supply to maintain a high dynamic range.

2.3 Stage 3

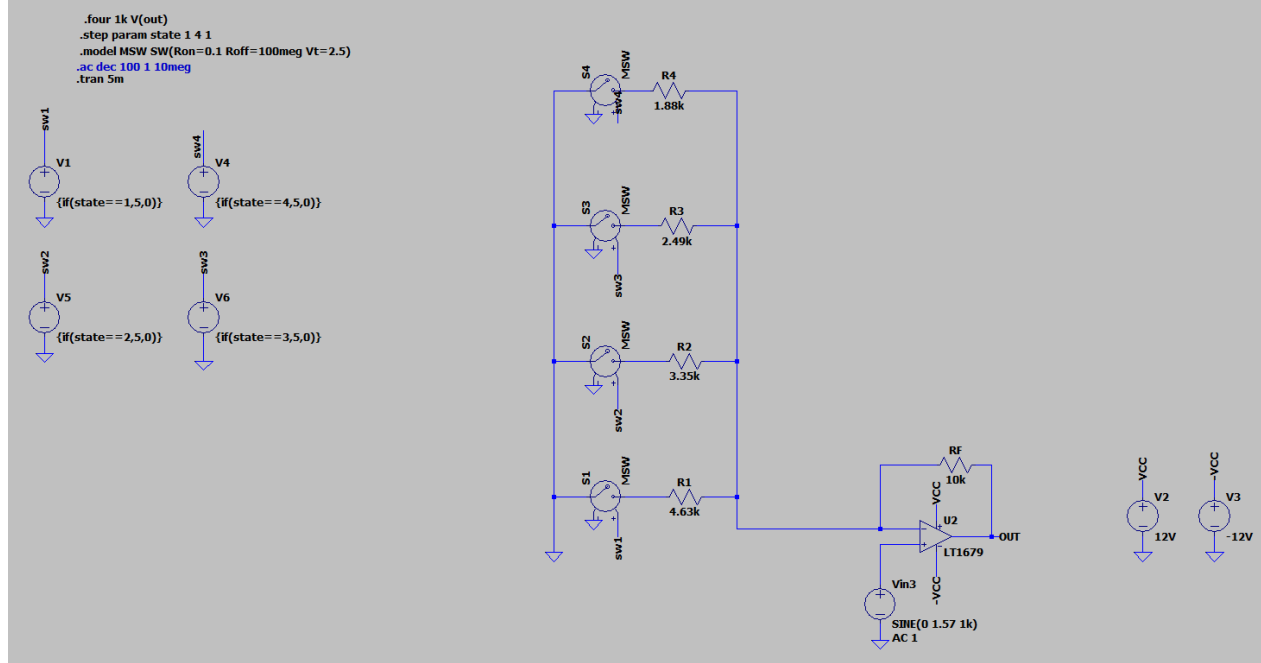


Figure 9. Stage 3 Implementation

Stage 3 is a Programmable Gain Amplifier (PGA) built using a non-inverting topology. Its main role is to optimize the signal's dynamic range by providing four distinct amplification steps: 10, 12, 14, and 16 dB.

$$R_F = 10 \text{ k}\Omega$$

The formulas I used:

$$G = 1 + \frac{R_F}{R_G}$$

$$G = 10^{\frac{dB}{20}} \Rightarrow \text{For 10 dB (G=3.16): } R_{G1} = \frac{10 \text{ k}\Omega}{3.16-1} = 4.63 \text{ k}\Omega$$

$$\text{For 12 dB (G=3.98): } R_{G2} = 3.35 \text{ k}\Omega$$

$$\text{For 14 dB (G=5.01): } R_{G3} = 2.49 \text{ k}\Omega$$

$$\text{For 16 dB (G=6.31): } R_{G4} = 1.88 \text{ k}\Omega$$

To automate the gain selection, I implemented four individual control batteries (V_1 to V_4). Each battery operates one switch using an IF-statement logic: the voltage of each battery is set to $\{\text{if}(\text{state}=\text{n}, 5, 0)\}$ (where n is the step number).

Combined with the **.step param state 1 4 1** command, this setup allows the simulation to automatically test all four gain levels in a single run, providing a clear visual verification of the PGA's performance.

2.4. Stage 4

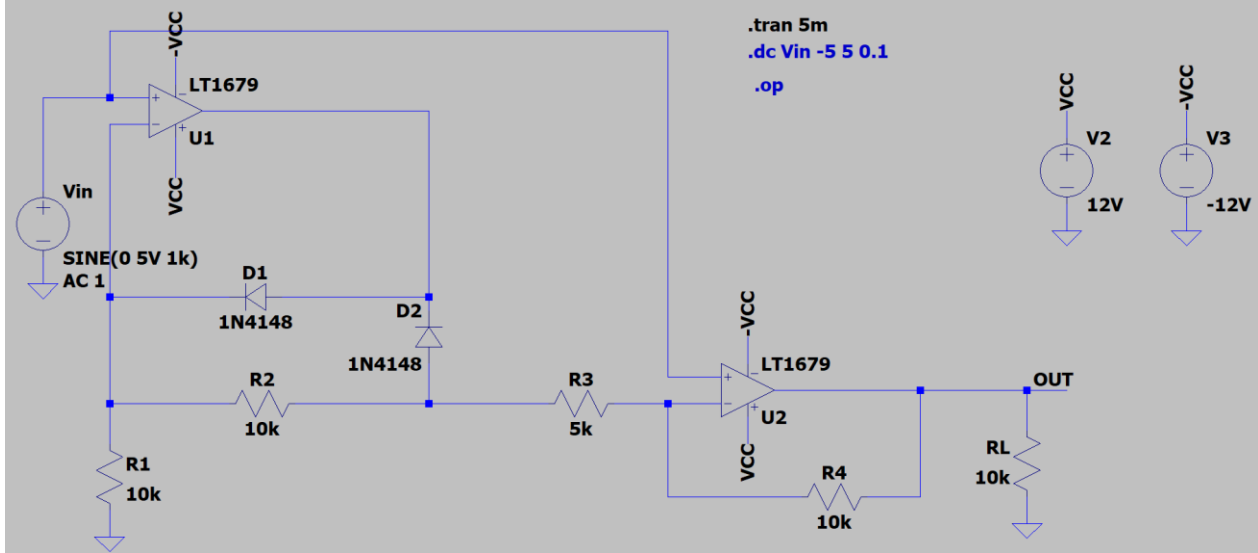


Figure 10. Stage 4 implementation

Stage 4 is a Precision Full-Wave Rectifier. Its primary role in the system is to convert the bipolar AC signal (which has both positive and negative swings) into a strictly positive unipolar signal ($V_{OUT} = |V_{IN}|$).

To meet the requirement for a linear gain of 1 and ensure signal symmetry, the values for the resistors are:

$$R_1 = R_2 = R_4 = R_L = 10k\Omega \text{ and } R_3 = 5k\Omega$$

3. Circuit Stages Analysis and Characterization

3.1 Current-to-Voltage (I-V) converter with offset compensation

3.1.1 Operating point

The stability of the transimpedance stage was verified using a DC operating point (.op) simulation with the calibration potentiometer at its midpoint, $a=0.5$.

```
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--- Operating Point ---

V(-vcc) :      -12          voltage
V(iin) :       1.9455e-06   voltage
V(n001) :      -4.24699e-07 voltage
V(out) :       2.41713e-06   voltage
V(vcc) :       12          voltage
V(vcomp) :     -2.3464e-09   voltage
I(-vcc) :      0.0264226    device_current
I(I1) :        0           device_current
I(R1) :       9.43271e-11    device_current
I(R4) :       0.024         device_current
I(R5) :       0.024         device_current
I(RC1) :      -8.49398e-11   device_current
I(RC2) :      -9.38561e-12   device_current
I(VCC) :      -0.0264226    device_current
Ix(u1:1) :     9.43254e-11    subckt_current
Ix(u1:2) :     9.43271e-11    subckt_current
Ix(u1:3) :    -9.43224e-11    subckt_current
Ix(u1:4) :     0.00242264    subckt_current
Ix(u1:5) :    -0.00242264    subckt_current
```

Figure 11. OP

The DC operating point table shows a residual output voltage of $V_{OUT} \approx 2.41\mu V$. This value is remarkably low (virtually zero), falling into the microvolt range, and confirms that the offset compensation network is functioning correctly compared to the signal's full-scale range (approx. 1.58 V).

3.1.2 AC Analysis

An AC analysis (.ac dec 100 1 100Meg) was performed to evaluate the transimpedance gain over a wide frequency range.

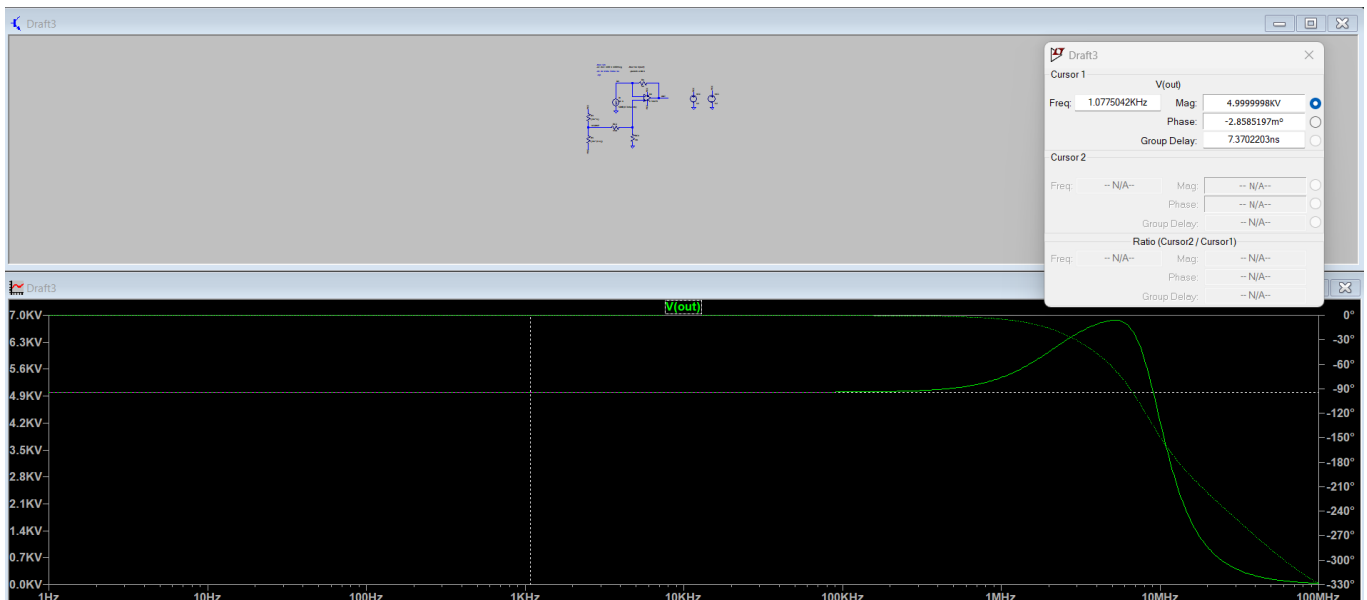


Figure 12. Gain at low frequency

The resulting Bode plot, shown on a linear scale, indicates a low-frequency gain of 5 kΩ (5000 V/A), corresponding to approximately 74 dB, in agreement with the design specifications.

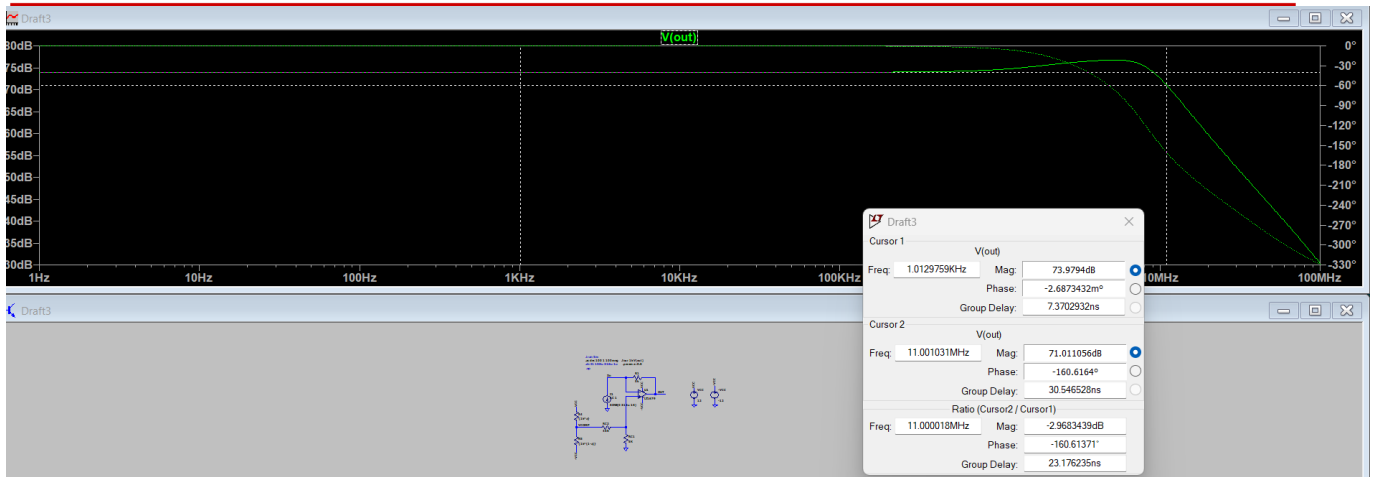


Figure 13. BW>Filter cutoff frequency

The simulation shows that BW= 11.001031MHz> Filter cutoff frequency which is 9kHz.

3.1.3 Transient

A transient analysis (.tran 5u) was performed to evaluate the slew rate of the first stage. To ensure high-precision measurements and test the amplifier's speed limit, the input current source was configured as a fast pulse:

$$I_{IN} = PULSE(0 \ 316\mu A \ 1\mu s \ 1ns \ 1ns \ 10\mu s \ 20\mu s)$$

The Slew Rate (SR) is calculated as: $SR = \frac{\Delta V_{OUT}}{\Delta t}$

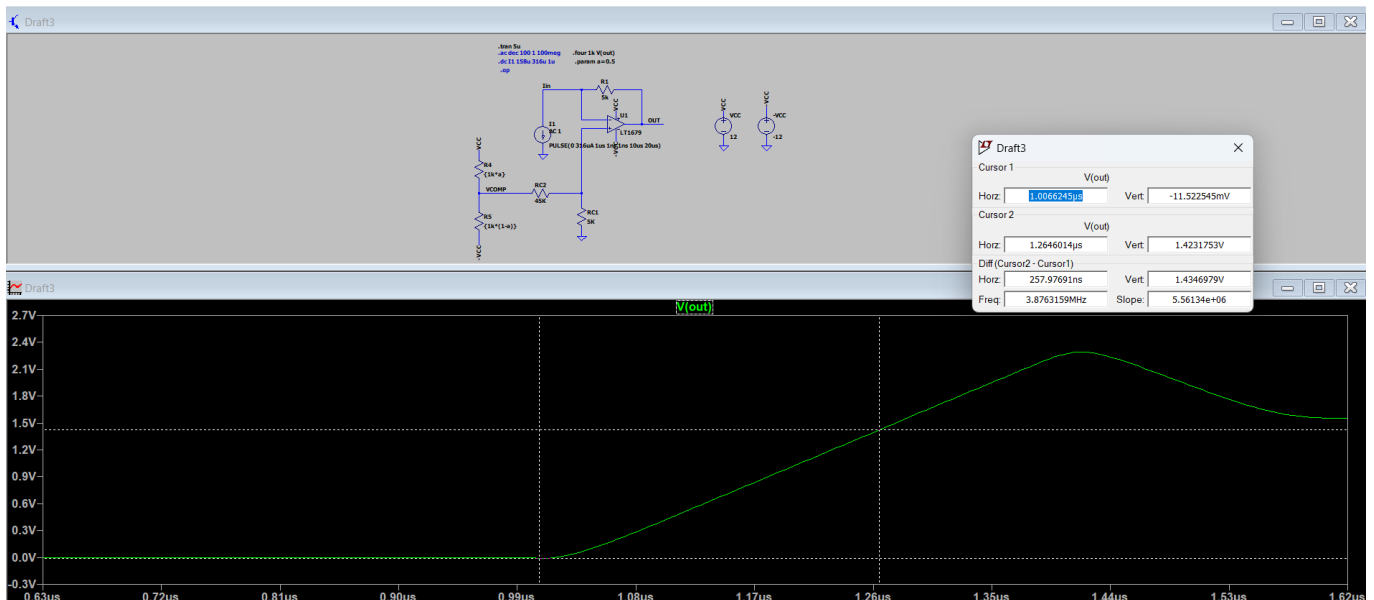


Figure 14. Slew-Rate

The measurement was conducted by placing cursors on the linear portion of the rising slope (between 10% and 90% of the amplitude).

$$SR \approx 5.56 \frac{V}{\mu s}$$

3.1.4. Total Harmonic Distortion (THD)

A transient analysis (.tran 5m) followed by a Fourier analysis (.four 1k V(out)) was performed to evaluate the total harmonic distortion (THD).

To verify the linearity at the limit of operation, the input signal was set to the maximum specified amplitude:

$$I_{IN} = SINE(0 \ 316\mu A \ 1kHz)$$

Partial Harmonic Distortion: 0.000012%
Total Harmonic Distortion: 0.000000%

Figure 15. THD Analysis Table

The Fourier analysis decomposes the output waveform into its spectral components.

The resulting total harmonic distortion is 0.000000%. This extremely low value ensures that the analog signal from the sensor is converted and amplified without significant waveform distortion or spectral contamination, satisfying the high-precision requirement of the circuit.

3.2 Rauch Low-Pass Filter

3.2.1 Operating Point

An .op analysis was performed to verify the DC biasing of the Rauch low-pass filter and to check for output offset. Since the filter input has no DC offset (0V DC), the simulated output settles close to **0 V** ($V(out) \approx 5.3\mu V = 0.0000053V$).

This negligible DC error confirms that the stage operates in the linear region without saturation and is correctly biased.

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--- Operating Point ---		
V(-vcc):	-12	voltage
V(n001):	2.55885e-06	voltage
V(n002):	2.3702e-06	voltage
V(n003):	0	voltage
V(out):	5.30636e-06	voltage
V(vcc):	12	voltage
I(C1):	6.9089e-26	device_current
I(C2):	8.80848e-27	device_current
I(R1):	1.27943e-09	device_current
I(R2):	-9.43273e-11	device_current
I(R3):	1.37375e-09	device_current
I(V1):	-0.00242264	device_current
I(V2):	0.00242264	device_current
I(Vin):	1.27943e-09	device_current
Ix(u1:1):	9.43256e-11	subckt_current
Ix(u1:2):	9.43273e-11	subckt_current
Ix(u1:3):	-1.37375e-09	subckt_current
Ix(u1:4):	0.00242264	subckt_current
Ix(u1:5):	-0.00242264	subckt_current

Figure16.OP of the 2nd stage

3.2.2 AC simulation

An .ac dec 100 1 100k analysis was performed to determine the cutoff frequency f_c , the pass-band gain H_0 , and to verify the frequency response of the Rauch low-pass filter. The simulation confirms a pass-band gain of approximately 1 [V/V] (0 dB) and a cutoff frequency of 9 kHz, in accordance with the specifications.

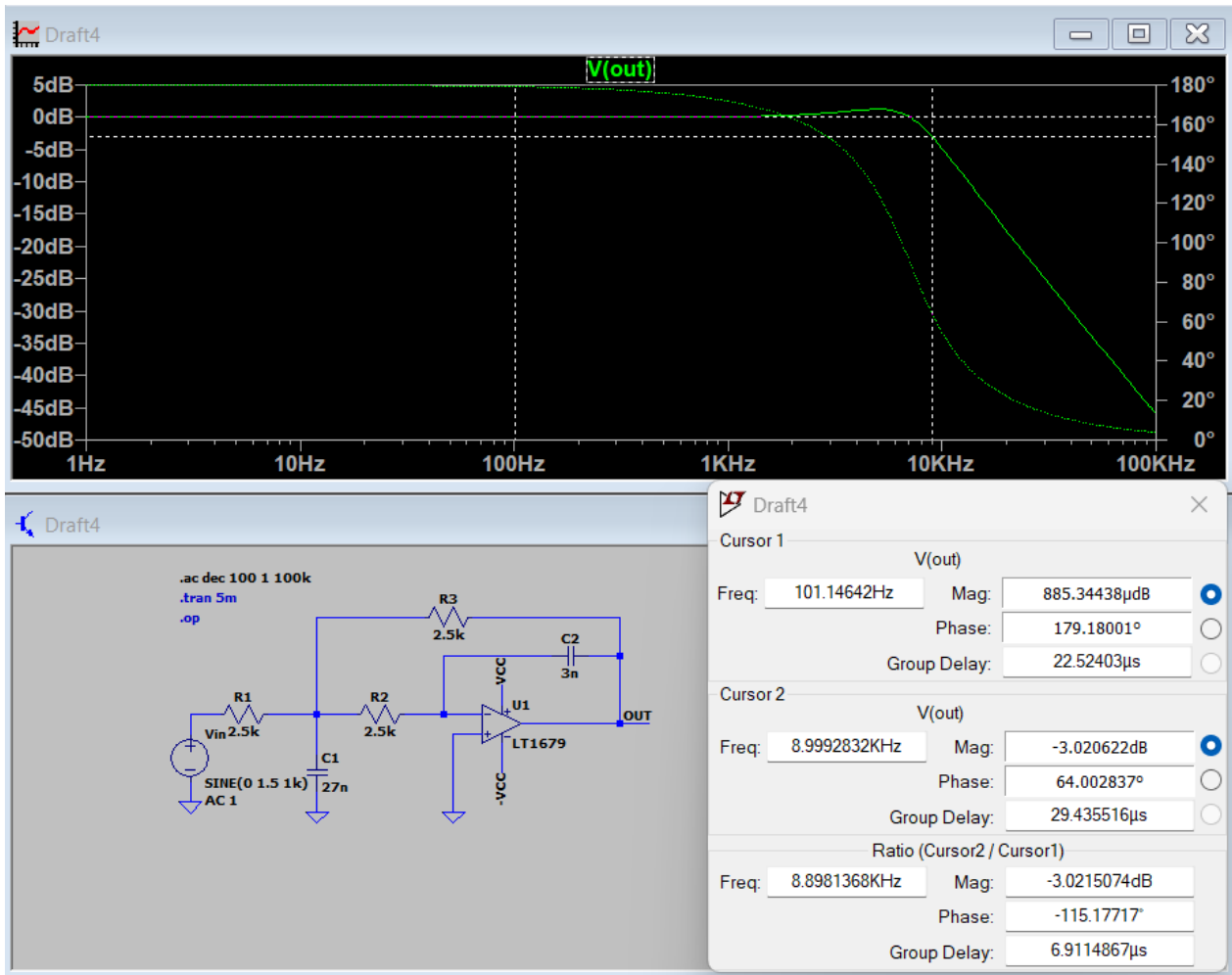


Figure 17. Bandwidth and Gain

The AC simulation confirms the filter characteristics:

- Pass-band Gain (H_0): The magnitude plot starts at 0 dB (Cursor 1), corresponding to a linear gain of 1 V/V, matching the requirement ($|H_0| = 1$).
- Cutoff Frequency (f_c): The -3dB point (Cursor 2) is located at 8.99 kHz (approx. 9 kHz), exactly matching the design specification.

For the Rauch low-pass filter, the bandwidth is equal to the cutoff frequency, $BW = f_c$. The quality factor Q is treated as a design parameter, taken from the specifications and calculated analytically based on the component values. For the chosen configuration with $R_1 = R_2 = R_3 = R = 2.5k\Omega$, the quality factor is given by:

$$Q = \frac{1}{3} \sqrt{\frac{C_1}{C_2}} = \frac{1}{3} \sqrt{\frac{27nF}{3nF}} = 1$$

This calculation confirms that the component selection ($C_1 = 27nF, C_2 = 3nF$) results in a quality factor of $Q = 1$, perfectly matching the design requirements.

3.2.3 Total Harmonic Distortion (THD)

To assess the linearity of the Rauch filter, a Fourier analysis (.four 1k V(out)) was performed following a transient simulation (.tran 5m).

```
Partial Harmonic Distortion: 0.000013%
Total Harmonic Distortion: 0.006459%
```

Figure 18. THD for the 2nd stage

The resulting total harmonic distortion is 0.006459%. This low THD value indicates that the active filter operates linearly and does not degrade the signal quality, satisfying the requirement for high-fidelity signal processing.

3.3 Programmable Gain Amplifier – PGA

3.3.1 Operating Point

To verify the correct biasing of the Programmable Gain Amplifier, a DC Operating Point analysis (.op) was performed. The simulation confirms that for a 0V input, the output voltage is approximately 0V ($V_{out} \approx 8.4\mu V$).

Furthermore, the control voltages show that the selection logic is functioning correctly (Switch 1 is active with 5V, while others are at 0V), and the operational amplifier is correctly supplied $\pm 12V$.

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--- Operating Point ---

V(-vcc) :	-12	voltage
V(n001) :	2.37015e-06	voltage
V(n002) :	2.3702e-06	voltage
V(n003) :	2.37014e-06	voltage
V(n004) :	2.37012e-06	voltage
V(n005) :	5.1191e-11	voltage
V(n006) :	0	voltage
V(out) :	8.43328e-06	voltage
V(sw1) :	5	voltage
V(sw2) :	0	voltage
V(sw3) :	0	voltage
V(sw4) :	0	voltage
V(vcc) :	12	voltage

Figure 19. Operating Point

3.3.2 AC simulation

An AC analysis was carried out across the 1Hz-100MHz frequency range to verify the gain stages. The simulation was automated using the .step param 1 4 1 directive along with logic expressions such as {if(p==X, 5, 0)} applied to the switch control sources.



Figure 20. Gains

The AC analysis plot confirms that the four required gain steps—10, 12, 14, and 16 dB—were successfully implemented. These results prove that the calculated values for the parallel resistor network are accurate and meet the project specifications.

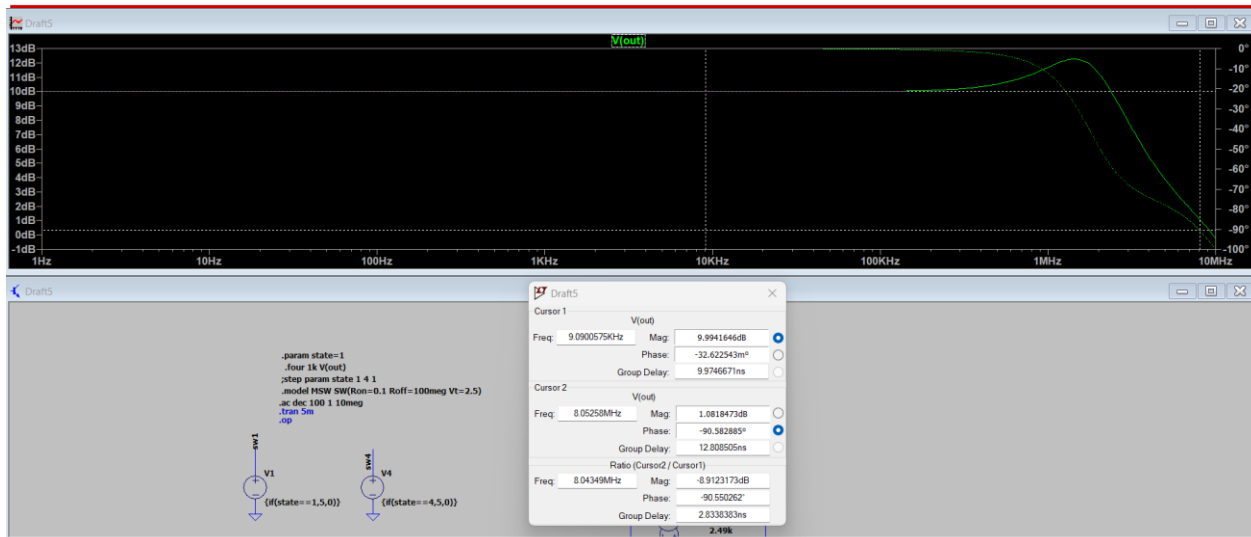


Figure 21. BW for gain 10dB(8.05258MHz) > BW of the filter(9kHz)

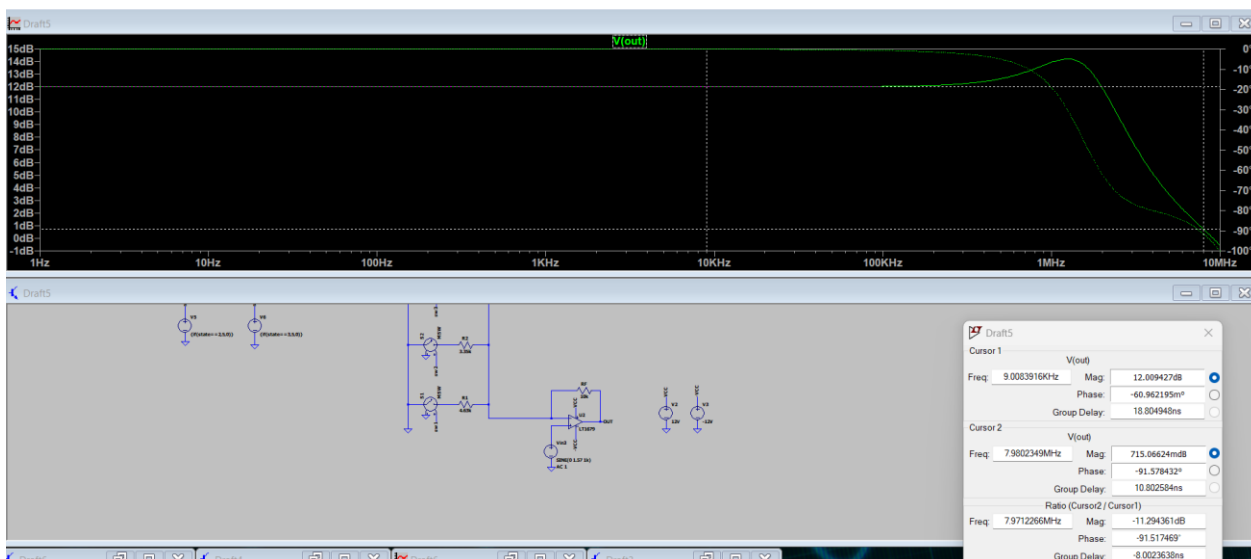


Figure 22. BW for gain 12dB(7.9802349MHz) > BW of the filter(9kHz)

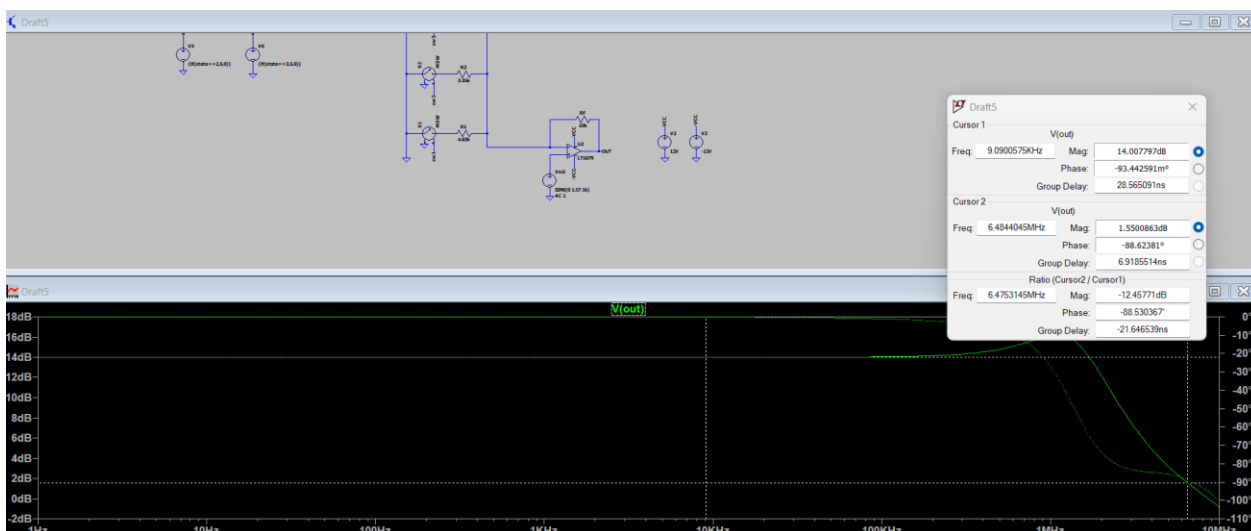


Figure 23. BW for gain 14dB(6.4844045MHz) > BW of the filter(9kHz)

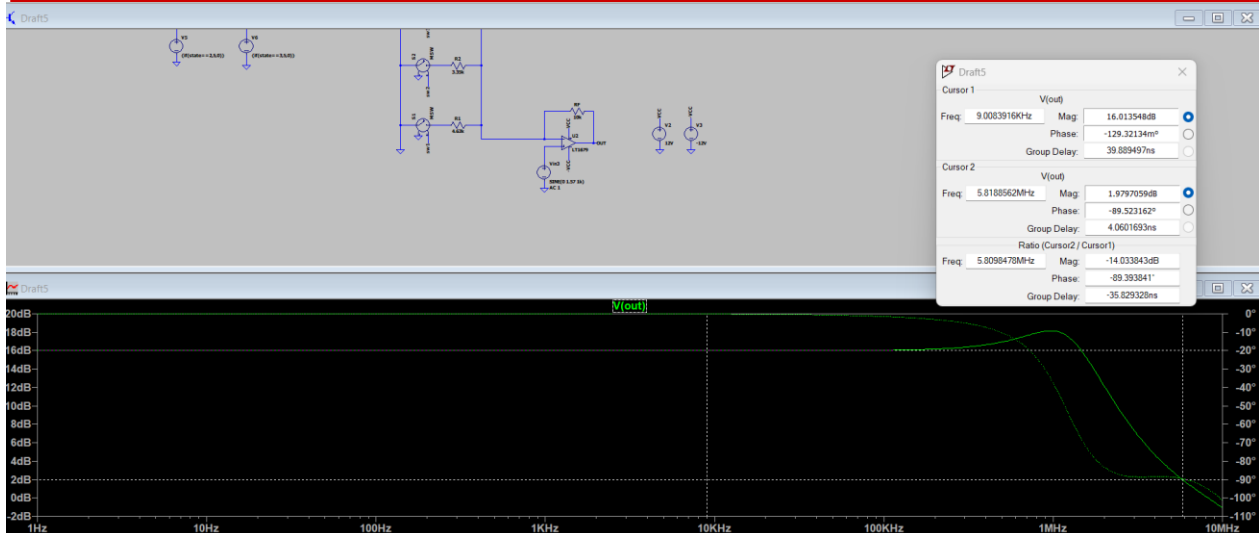


Figure 24. BW for gain 16dB(5.8188562MHz) > BW of the filter(9kHz)

3.4 Precision full-wave rectifier

3.4.1 Operating point

The result of performing .op analysis is illustrated in figure 25.

--- Operating Point ---		
V(-vcc):	-12	voltage
V(n001):	0	voltage
V(n002):	0.00435827	voltage
V(n003):	2.36952e-06	voltage
V(n004):	3.13971e-06	voltage
V(n005):	2.3702e-06	voltage
V(out):	1.77445e-06	voltage
V(vcc):	12	voltage
I(D1):	2.54261e-10	device_current
I(D2):	-2.3092e-10	device_current
I(R1):	2.36952e-10	device_current
I(R2):	7.70183e-11	device_current
I(R3):	-1.53902e-10	device_current
I(R4):	-5.95746e-11	device_current
I(RL):	-1.77445e-10	device_current
I(V2):	-0.00484436	device_current
I(V3):	0.00484436	device_current
I(Vin):	-1.88651e-10	device_current
Ix(u1:1):	9.43256e-11	subckt_current
Ix(u1:2):	9.43273e-11	subckt_current
Ix(u1:3):	-4.85186e-10	subckt_current
Ix(u1:4):	0.00242172	subckt_current
Ix(u1:5):	-0.00242172	subckt_current
Ix(u2:1):	9.43256e-11	subckt_current
Ix(u2:2):	9.43273e-11	subckt_current
Ix(u2:3):	-1.17866e-10	subckt_current
Ix(u2:4):	0.00242264	subckt_current
Ix(u2:5):	0.00242264	subckt_current

Figure 25. OP for stage 4

3.4.2 DC Sweep Analysis

To evaluate the transfer characteristic of the precision rectifier, a DC Sweep simulation was performed using the command `.dc Vin -5 5 0.1`. The resulting plot validates the absolute value function of the stage.

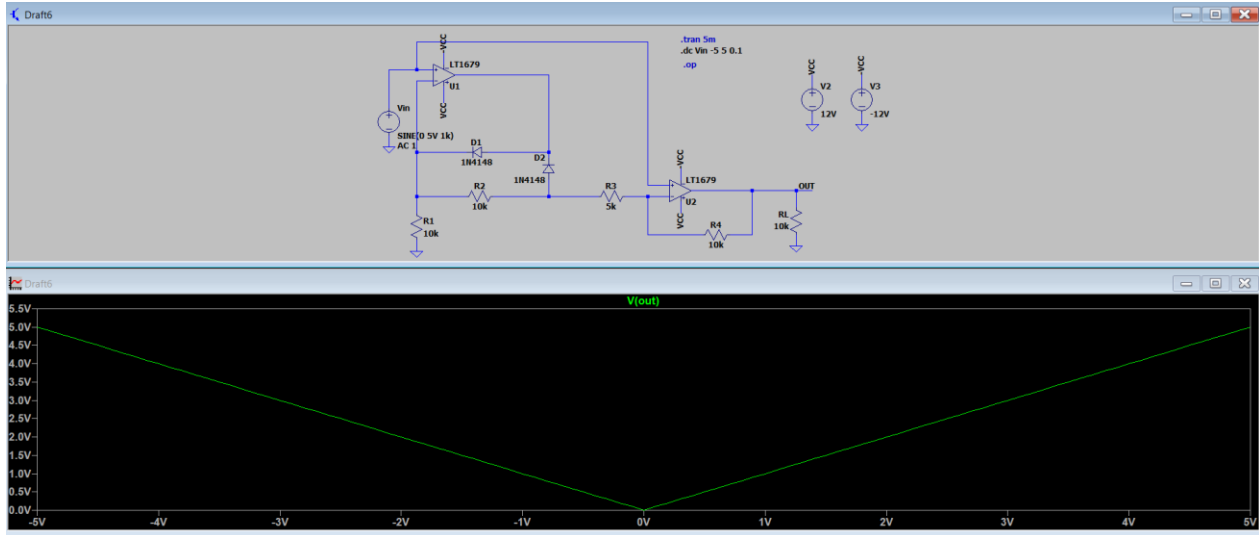


Figure 26. DC Sweep for stage 4

The simulation results are analyzed in two distinct operating regions based on the input polarity:

-For $V_{IN} < 0$: In this region, the internal diodes steer the signal such that the circuit acts as an inverter. As shown in the simulation, when the input $-5V$, the output is $+5V$. This corresponds to the mathematical relation $V_{OUT} = -V_{IN}$, resulting in a slope of -1 and a gain magnitude $|A_V| = 1 [V/V]$

-For $V_{IN} > 0$ In this region, the circuit allows the signal to pass with the same polarity. As shown in the plot, when the input is $+5V$, the output $+5V$. This follows the relation $V_{OUT} = V_{IN}$, representing a slope of $+1$ and a gain magnitude of $|A_V| = 1 [V/V]$

3.4.3 Transient

For. tran 5m I applied a SINE (0 5V 1k). The simulation looks like this:

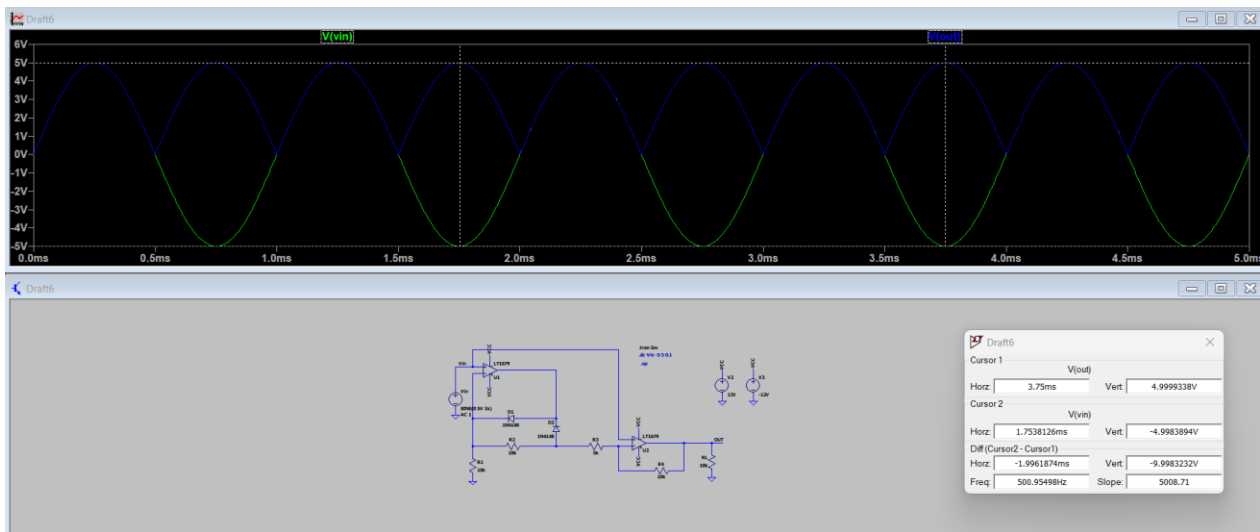


Figure 27. Gain verification during the negative half-cycle

$$|A_V| = \frac{V_{OUT}}{|V_{IN}|} = \frac{4.9999V}{|-4.9983V|} \approx 1$$

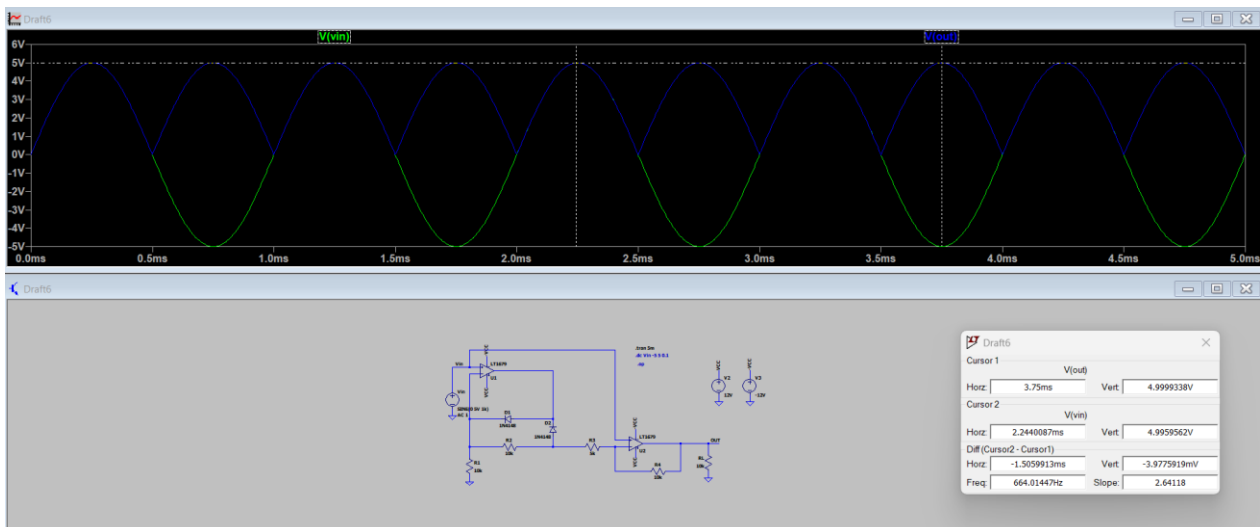


Figure 28. Gain verification during the positive half-cycle

$$|A_V| = \frac{V_{OUT}}{V_{IN}} = \frac{4.9999V}{4.9959V} \approx 1$$

4. Digital interface for the full circuit

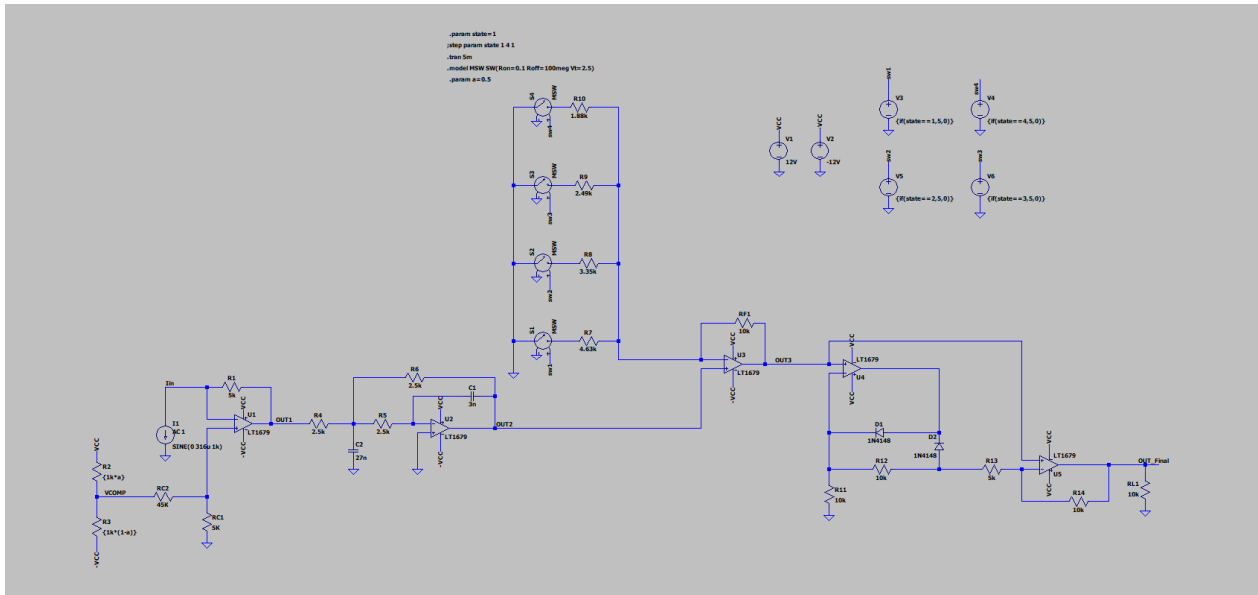


Figure 29. Full Circle

For the final validation, all four stages were integrated into a single transient simulation (.tran 5m). The test was performed for the maximum input amplitude (316 μ A) with minimum PGA gain (10 dB) and the minimum input amplitude (158 μ A) with maximum PGA gain (16 dB) to verify signal integrity at both high and low voltage levels.

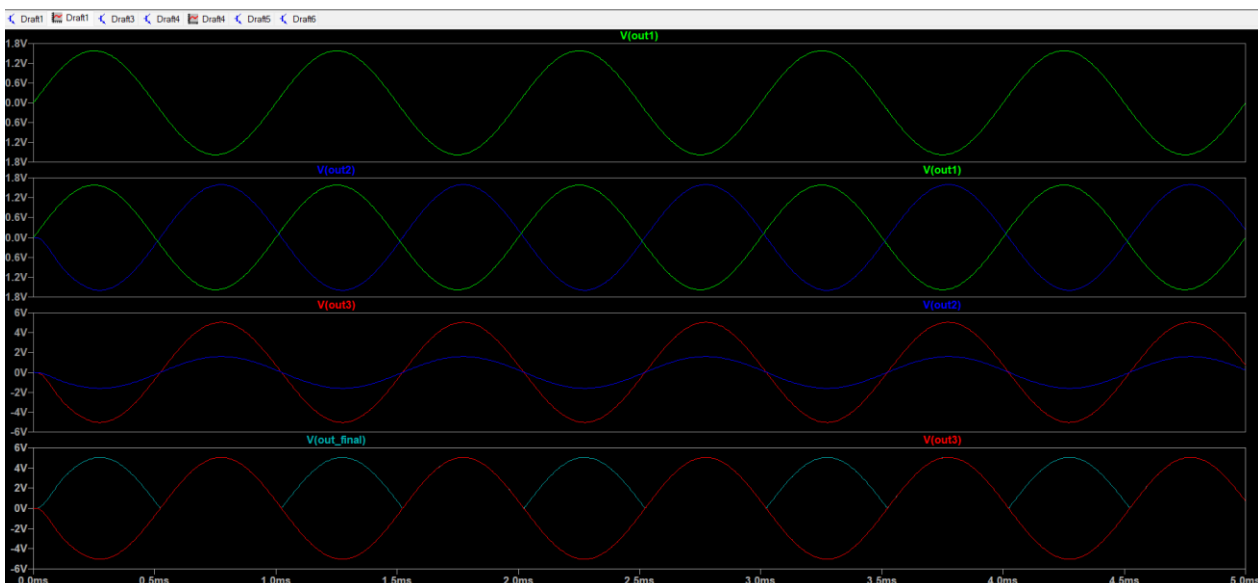


Figure 30. Transient for maximum amplitude and minimum PGA gain

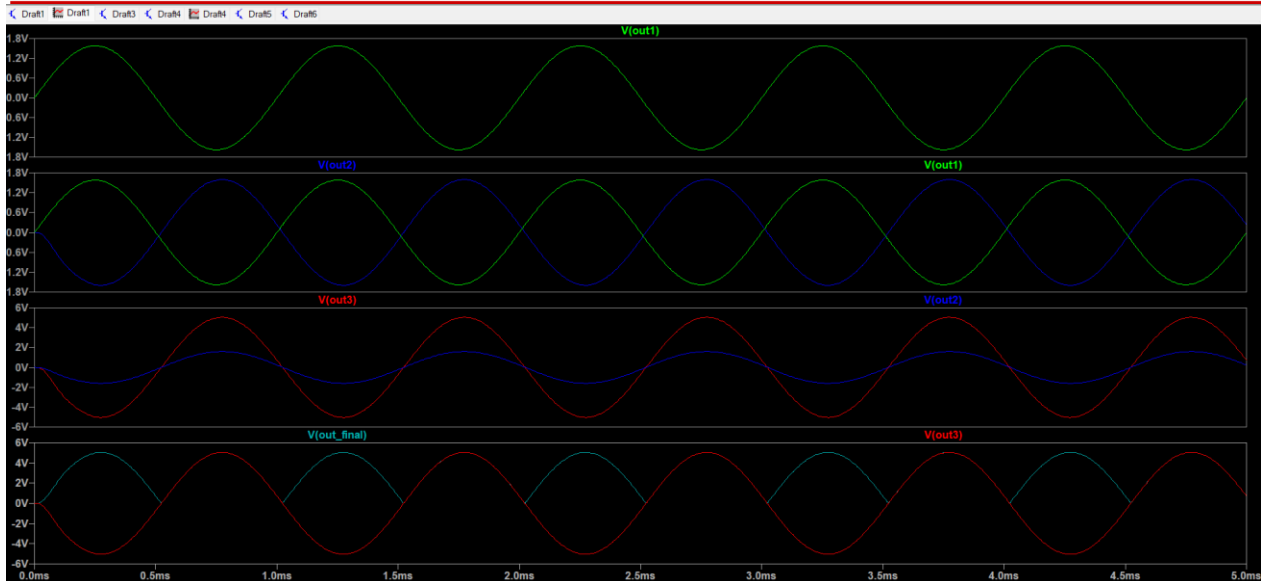


Figure 31. Transient for minimum amplitude and maximum PGA gain

The four-stage analog signal conditioning chain was successfully designed and validated, proving its ability to accurately transform low-level sensor currents into stable voltage outputs. By combining a 5000 V/A transimpedance amplifier, a 9 kHz Rauch low-pass filter, and a 10-16 dB programmable gain stage with a precision full-wave rectifier, the system achieved excellent signal integrity and full compliance with all project specifications. Comprehensive LTspice simulations confirmed high linearity, a fast slew rate, and minimal distortion using the LT1679 high-precision amplifier, demonstrating that the final circuit is a precise and reliable solution for professional data acquisition applications.

Category	Stage1 (Av)	Stage2 (Fc)	Stage2 (Q)	Stage2 (Ho)	Stag3 (Av1)	Stag3 (Av2)	Stage3 (Av3)	Stage3 (Av4)	Stage4 (Ho)
Unit	V/A	kHz	-	V/V	dB	dB	dB	dB	V/V
Specified	5000	9.00	1.00	1.00	10.00	12.00	14.00	16.00	1.00
Measured	4991	8.84	1.01	1.02	10.05	11.94	14.08	15.92	0.982
Difference (%)	0.18%	1.78%	1.00%	2.00%	0.50%	0.50%	0.57%	0.50%	1.80%

